

# CONNECTION DESIGN ADVISOR

Project: Quick Analysis

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Of course. As STRUCTMIND AI, I will provide optimal connection recommendations for your 3-story SMF + OCBF mixed system.

This analysis is based on the requirements of AISC 360-22 (Specification for Structural Steel Buildings) and, critically, AISC 341-22 (Seismic Provisions for Structural Steel Buildings), which governs the design of seismic force-resisting systems (SFRS) like SMF and OCBF.

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## A. Beam-Column Joints (Special Moment Frame - SMF)

These connections are the most critical for seismic performance, designed to be ductile and form plastic hinges in the beams, not the columns or the connections themselves.

### 1. Recommended Connection Type:

- Reduced Beam Section (RBS) or "Dogbone" Connection. This is a highly reliable and widely used prequalified moment connection per AISC 358-22. The flanges of the beam are trimmed in a specific radius cut, creating a weakened section that acts as the ductile "fuse" or plastic hinge, protecting the column and the beam-to-column weld.

### 2. Governing Limit States (per AISC 341-22, Section E3 & AISC 360-22, Chapter J):

- Design Philosophy: The connection must be designed to be stronger than the beam. The required strength is based on developing the probable maximum moment at the plastic hinge,  $M_{pr}$ , which accounts for material overstrength ( $R_y$ ) and strain hardening ( $C_{pr}$ ).
- Beam: Flexural Yielding (Plastic Hinge Formation) at the RBS location. (AISC 360, F2)
- Beam-to-Column Connection (CJP Weld & Bolted Shear Tab):
  - The CJP flange welds and web connection must be designed for the shear corresponding to  $M_{pr}$  at the column face.
  - Weld Rupture: The demand-critical CJP groove weld strength must exceed the demand. (AISC 360, J2.4)
  - Bolt Shear Rupture for the shear tab bolts. (AISC 360, J3.7)
  - Bolt Bearing/Tearout on the beam web and shear tab. (AISC 360, J3.10)

- Block Shear Rupture of the beam web at the shear tab. (AISC 360, J4.3)

- Column:

- Panel Zone Shear Yielding: The column web panel zone must be checked for shear forces from the beam moments. Doubler plates may be required. (AISC 341, E3.6e)

- Strong Column-Weak Beam (SCWB): The sum of column flexural strengths must exceed the sum of beam flexural strengths at the joint. (AISC 341, E3.4a)

### 3. Bolt / Weld Sizing Guidance:

- Bolts: Use high-strength bolts (e.g., ASTM F3125 Gr. A325 or A490) for the shear tab. Size them for the shear generated when the RBS reaches its full plastic capacity.

- Welds: The beam flange-to-column flange connection must be a Complete Joint Penetration (CJP) groove weld, designated as "demand critical" per AISC 341. This requires stringent quality control and inspection (UT testing).

### 4. Constructability Score: 9/10

- Remarks: Excellent. The complex work (RBS cutting, CJP welding) is done in the controlled environment of the fabrication shop. Field work is reduced to simple bolting of the shear tab, making erection fast and reliable.

### 5. Alternate Solutions:

- Bolted Flange Plate (BFP): Good alternative, also prequalified in AISC 358. Involves field bolting of the flanges. More field labor and inspection (slip-critical bolt tensioning) but avoids field welding.

Constructability Score: 7/10.

- Bolted End Plate (BEP): Can be very efficient but requires tighter fabrication and erection tolerances for plumb and fit-up.

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## **B. Brace-to-Gusset Connection (Ordinary Concentrically Braced Frame - OCBF)**

This connection must accommodate brace buckling and tensile yielding. The design is governed by capacity-based principles.

### 1. Recommended Connection Type:

- Gusset Plate Connection. For HSS braces, this typically involves welding the HSS directly to the gusset plate in the shop and bolting the gusset to the beam/column in the field.

### 2. Governing Limit States (per AISC 341-22, Section F1 & AISC 360-22, Chapter J):

- Design Philosophy: The connection strength is based on the expected strength of the brace member.

- Required Tensile Strength =  $R_y F_y A_g$  (Expected Yield Strength)

- Required Flexural/Post-Buckling Strength =  $1.1 R_y P_n$  (Expected Post-Buckling Strength) for some checks or as specified.
- Gusset Plate:
  - Whitmore Section Yielding/Rupture: The effective width of the gusset must be able to develop the brace tensile strength. (AISC 360, J4.4)
  - Block Shear Rupture at the bolted interface with the frame. (AISC 360, J4.3)
  - Gusset Buckling: The unsupported edges of the gusset must be checked for buckling. (AISC Design Guide 29)
- Brace to Gusset Weld:
  - Weld Rupture: The weld group must be sized to develop the full tensile yielding strength of the brace. (AISC 360, J2.4)
- Gusset to Frame Bolts:
  - Bolt Shear Rupture and Bearing/Tearout. (AISC 360, J3.7 & J3.10)

### 3. Bolt / Weld Sizing Guidance:

- Bolts: ASTM F3125 Gr. A325 bolts are typically sufficient. Size for the required tensile strength of the brace.
- Welds: Use fillet welds. The total weld capacity around the HSS perimeter must exceed  $R_y F_y A_g$  of the brace.
- Detailing: Detail the gusset plate to allow for out-of-plane rotation as the brace buckles, typically by providing a clearance (2x plate thickness) at the end of the brace.

### 4. Constructability Score: 8/10

- Remarks: Very common and well-understood connection. Profiling the HSS to fit the gusset is a standard shop practice. Field erection is straightforward bolting.

### 5. Alternate Solutions:

- Bolted End Tee/Angle: A structural tee or double angles can be welded to the HSS brace in the shop, allowing for a fully bolted field connection. This can simplify fit-up if brace lengths vary.

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## C. Column Splices

Column splices are critical for load transfer and erection efficiency. Their location and strength are governed by seismic requirements.

### 1. Recommended Connection Type:

- Bolted Flange Plate Splice. This is the most common and constructible type for field splices. Shear can be transferred through bearing plates or a supplemental web splice.

#### 2. Governing Limit States (per AISC 341-22, E3.6f & AISC 360-22, J1):

- Location: Splices must be located at least 4 ft away from beam-column joints. The ideal location is near the point of contraflexure (inflection point), roughly at mid-story height.
- SMF Columns: The splice must develop the shear and moments from load combinations. It must have a required flexural strength of at least  $0.5 R_y M_p$  of the smaller column section.
- OCBF Columns: Designed for the required strength from load combinations including amplified seismic load ( $\Omega_o$ ).
- Limit States:
  - Bolt Shear Rupture and Bearing/Tearout on flanges and plates. (AISC 360, J3.7 & J3.10)
  - Gross Section Yielding and Net Section Rupture of the splice plates. (AISC 360, J4.1)

#### 3. Bolt / Weld Sizing Guidance:

- Bolts: Use high-strength, slip-critical bolts (Class B faying surface) to prevent slip under load reversal and provide rigidity.
- Plates: Plate thickness is driven by the net section rupture limit state to transfer the flange force.

#### 4. Constructability Score: 9/10

- Remarks: Highly constructible and safe for erection. Using erection aids like a seat angle welded to the lower column shaft is recommended. Allows for tolerance in column length.

#### 5. Alternate Solutions:

- Complete Joint Penetration (CJP) Welded Splice: Provides a seamless load path but is very expensive, slow, and difficult to execute and inspect in the field. Generally avoided unless architecturally required.

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## D. Column Base Plates

The base plate anchors the entire structure and must transfer all forces (axial, shear, moment) into the foundation.

#### 1. Recommended Connection Type:

- Exposed Column Base Plate. Standard design using a thick base plate, anchor rods, and shear lugs if required.

#### 2. Governing Limit States (per AISC 341-22, D2.6 & AISC 360-22, J8):

- SMF Columns: The base connection must be designed to remain essentially elastic and force the plastic hinge to form in the column just above the base. The required strength is determined by the probable

column flexural strength,  $M_{pc}$ .

- OCBF Columns: Designed for the required strength from load combinations including amplified seismic load ( $\Omega_o$ ).
- Limit States:
- Concrete Bearing/Crushing under the base plate. (AISC 360, J8)
- Base Plate Bending (Yielding) at the flange and web locations. (AISC Design Guide 1)
- Anchor Rod Tensile/Shear Rupture: Based on ACI 318, Chapter 17.
- Concrete Breakout (Tension and Shear) for the anchor group.
- Weld Rupture for column-to-base plate weld.

### 3. Bolt / Weld Sizing Guidance:

- Anchor Rods: Use ductile steel grades (ASTM F1554 Gr. 36 or 55). Grade 105 is less ductile and should be used with caution. Size for moment-induced tension and total shear.
- Plate Thickness: Governed by plate bending. A thick plate is required to distribute column flange forces without yielding.
- Welds: Use CJP or double-sided fillet welds to attach the column to the base plate. Must develop the required column strength.

### 4. Constructability Score: 7/10

- Remarks: Standard practice, but precision is critical. Anchor bolt placement via template is crucial. Leveling nuts and non-shrink grout are standard procedure. Can be a bottleneck if not coordinated properly.

### 5. Alternate Solutions:

- Embedded Column Base: The column is embedded into the concrete foundation. Can be a very robust solution but complicates foundation forming and concrete placement.

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### Summary Remarks:

For this mixed system, the design is driven by the stringent ductility and capacity-design requirements of AISC 341-22. The SMF connections are designed to be the ductile elements, while all other components in the load path (gussets, base plates, splices) are designed to remain elastic by being stronger than the expected capacity of the "fuses." Careful detailing and quality control, especially for demand-critical welds, are paramount to achieving the intended seismic performance.